

A. Classification task

Embeddings. Table 5 details the parameters explored during the two phases: (1) Random search across a set of 500 parameters. (2) Full search on a subset of these parameters. Following the random search, we observed a significant negative correlation between the learning rate and the Mean Reciprocal Rank (MRR) score ($corr = -0.20$, $pval = 6.46 \times 10^{-6}$). Table 6 displays the optimal values identified for each model during the random search, highlighting that DistMult achieved the highest scores. Further investigation into results with a learning rate of 0.001 and DistMult as the model revealed that metrics improved slightly with larger embedding dimensions, fewer negative samples, and more epochs. These insights guided the parameters selected for the subsequent full search (2). After conducting the remaining 378 experiments, we found a strong negative correlation between the number of negative samples and the MRR score ($corr = -0.48$, $pval = 3.11 \times 10^{-23}$). Specifically, experiments with 1 negative sample showed a significant positive correlation between the number of epochs and the MRR score ($corr = -0.56$, $pval = 2.0 \times 10^{-6}$). No significant differences were found in embedding dimensions, thus we retained the dimension (336) that yielded the best score.

Table 5. Search space for the hyperparameter optimisation for TransE [50], DistMult [51], ComplEx [52], and R-GCN [53]. The learning include the following values: [0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1]. The left part describes the parameters for the random search with 500 parameters. After filtering, the right part describes the parameters for the final search with 378 parameters. The model we used was DistMult for this final search. Dim.: embedding dimension; lr: learning rate; # of negs: number of negatives; # of epochs: number of epochs.

Phase	1			2		
Hyperparam.	min	max	step	min	max	step
Dim.	16	512	16	192	512	16
lr	0.001	0.1	log	0.001	0.001	-
# of negs	1	100	10	1	60	10
# of epochs	100	500	100	300	500	100

Table 6. Optimal values identified for the random search.

Hyperparam.	TransE [50]	DistMult [51]	ComplEx [52]	R-GCN [53]
# configs	131	111	133	135
Embedding dim.	16	416	32	80
Learning rate	0.02	0.001	0.02	0.005
# of negatives	90	1	30	100
# of epochs	300	500	400	500
Hits@1	0.0957	0.215	0.0800	0.146
Hits@3	0.186	0.391	0.131	0.296
Hits@10	0.289	0.501	0.195	0.489
MRR	0.167	0.320	0.122	0.259

Model. Table 7 shows the optimal parameters we searched for the model: we trained the model on the test set and evaluated it on the validation set. Table 8 shows the best parameters for each hypothesis type.

Table 7. Hyperparameter search for the classification.

Parameter	Values
<i>criterion</i>	[gini, entropy]
<i>splitter</i>	[best, random]
<i>max_depth</i>	[None, 5, 15, 20]
<i>min_samples_split</i>	[2, 5, 10]
<i>min_samples_leaf</i>	[1, 2, 4]
<i>max_features</i>	[None, sqrt, log2, 0.5, 0.7]
<i>max_leaf_nodes</i>	[None, 10, 30]

Table 8. Best parameters for the classification task, with a train/val/test split of 0.8/0.1/0.1.

Hypothesis	Regular	Study-Mod	Var-Mod
Best Parameters			
<i>criterion</i>	gini	entropy	gini
<i>splitter</i>	random	random	random
<i>max_depth</i>	20	15	5
<i>min_samples_split</i>	5	10	2
<i>min_samples_leaf</i>	2	2	2
<i>max_features</i>	sqrt	None	0.5
<i>max_leaf_nodes</i>	None	None	None

Complex Models. We compared the performance of the simple decision tree classifier with more complex classification models. For each model we compared different hyperparameters. Table 9 shows the best accuracy on the validation set for each model, along with the corresponding accuracy on the train set and the number of experiments run. Given the similarity of results compared to the decision tree, we opted for simplicity and kept the decision tree classifier. All results related to this part can be found in the `experiments/classification/complex` folder of this repository.

Table 9. Mean ranks and scores for all hypotheses. A higher score is better, a lower rank is better. Bold: best metric, underlined: best second metric.

Model type	Model	reg			study-mod			var-mod		
		# exp.	Train	Val.	# exp.	Train	Val.	# exp.	Train	Val.
Tree	DecisionTree	-	0.79	0.78	-	0.80	0.80	-	0.74	0.70
NN	MLP	96	0.77	0.74	196	0.80	0.80	96	0.73	0.67
Neighbors	KNeighbors	72	0.79	0.72	72	0.81	0.75	72	0.71	0.77
Linear	Ridge				19	0.77	0.78			
Linear	SGDC	96	0.77	0.76	96	0.77	0.78	96	0.74	0.72
Ensemble	AdaBoost	9	0.75	0.76	9	0.77	0.78	9	0.74	0.70
Ensemble	Bagging	100	0.79	0.78	100	0.81	0.78	100	0.76	0.74
Ensemble	GradientBoosting	100	0.79	0.78	100	0.80	0.80	100	0.61	0.70
Ensemble	Random Forest	100	0.77	0.78	100	0.81	0.80	100	0.73	0.70

Table 10. Optimal values identified for the random search for the link prediction method. T: TransE [50], D: DistMult [51], C: ComplEx [52], R: R-GCN [53].

Hyperparam.	reg				study-mod				var-mod			
	T	D	C	R	T	D	C	R	T	D	C	R
# configs	14	9	9	8	14	9	9	8	14	9	9	8
Embedding Dim.	272	192	432	512	432	496	32	160	432	496	32	304
Learning rate	0.02	0.1	0.1	0.01	0.0	0.01	0.01	0.0	0.0	0.01	0.01	0.02
# of negatives	10	1	80	20	80	10	40	100	80	10	40	70
# of epochs	100	200	200	200	300	500	300	300	300	500	300	500
Hits@1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.04	0.0	0.0	0.0	0.0
Hits@3	0.01	0.0	0.0	0.0	0.01	0.0	0.0	0.11	0.02	0.0	0.01	0.02
Hits@10	0.01	0.0	0.01	0.0	0.03	0.0	0.01	0.16	0.05	0.02	0.03	0.02
MRR	0.01	0.0	0.01	0.0	0.01	0.0	0.0	0.08	0.02	0.0	0.01	0.01

B. Link Prediction task

Sub-symbolic methods. Table 10 shows the results of the best parameters for each model and hypothesis type on the validation set. Given the very low results on these models and despite the parameter optimisation, we decided to discard the models, especially since their symbolic counterpart performed much better.

Symbolic method (AnyBURL). Figure 7 shows the distribution of the support scores of the rules learned by AnyBURL. The top figure shows the percentage distribution and for the three types of hypothesis, the majority of the support scores are below 0.1. The bottom figure shows the count distribution having removed the scores below 0.1.

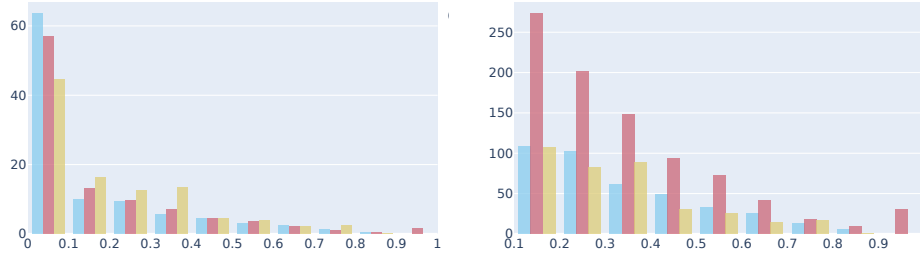
**Figure 7.** Distributions of support scores of rules learned by AnyBURL. Top: percentage distribution; bottom: count distributes for rules whose support score is above 0.1 for more clarity. Blue: reg, red: study-mod, yellow: var-mod.

Figure 8 illustrates the distribution of support scores for rules, categorized by the different antecedents on the x-axis. The scores are mostly between 0.2 and 0.4, and the rules with `cp:mod` as antecedents are the ones with the lowest support scores. Figure 9 shows the top 3 rules learned for each type of hypothesis. All these rules are of length 1. For instance, the top rule with a score of 0.85 for the var-mod rule states that studies with `id:uncertaintylevel/high` as the second `sivv` has a `positiveEffectOn` contributions.

January 2025

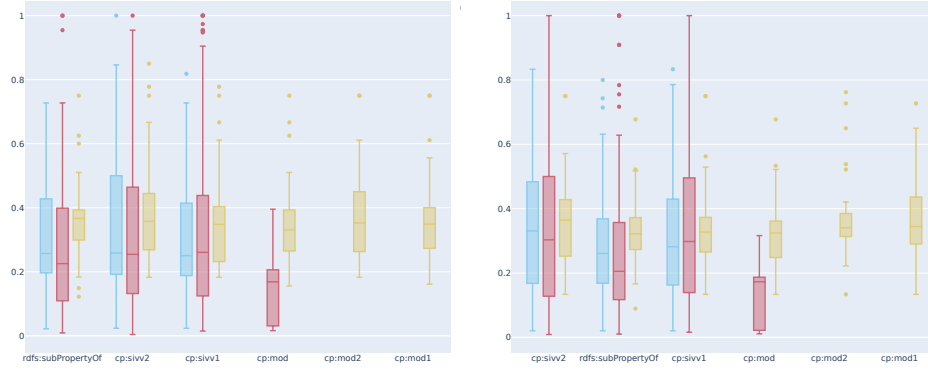


Figure 8. Boxplot of scores for antecedents in rules. Left: positive effect, right: negative effect. Blue: reg, red: study-mod, yellow: var-mod.

Top 3 Rules Learned for the reg hypothesis

```
cp:hasPositiveEffectOn(X,cooperation)
<= cp:sivv2(X,id:intergroupcomp/intergroup_prisoner's_dilemma)(1.0)
cp:hasPositiveEffectOn(X,cooperation)
<= cp:sivv2(X,id:motivationalorientation/competitive)(0.85)
cp:hasNegativeEffectOn(X,contributions)
<= cp:sivv1(X,id:uncertaintytarget/threshold)(0.83)
```

Top 3 Rules Learned for the study-mod hypothesis

```
cp:hasNegativeEffectOn(X,cooperation)
<= rdfs:subPropertyOf(X,cp:continuationProbabilityLevel)(1.0)
cp:hasPositiveEffectOn(X,contributions)
<= cp:sivv1(X,id:religiouslevel/high)(1.0)
cp:hasNegativeEffectOn(X,cooperation)
<= cp:sivv2(X,id:emotion/happiness)(1.0)
```

Top 3 Rules Learned for the var-mod hypothesis

```
cp:hasPositiveEffectOn(X,contributions)
<= cp:sivv2(X,id:uncertaintylevel/high)(0.85)
cp:hasPositiveEffectOn(X,cooperation)
<= cp:sivv2(X,id:decisionmaker/group)(0.78)
cp:hasPositiveEffectOn(X,cooperation)
<= cp:sivv1(X,id:decisionmaker/individual)(0.78)
```

Figure 9. Top 3 Rules Learned for the Three Types of Hypotheses.

C. LLM task

Figure 10, 11 and 12 shows examples of prompts we used for the reg, study-mod and var-mod hypotheses respectively. The only difference is in the data format and the

template hypothesis.

Prompt:

You will be given data in .csv format, where each line can be read as a hypothesis on how humans cooperate. The csv has the following columns: 'dependent', 'giv_prop', 'iv', 'iv_label', 'cat_t1', 'cat_t1_label', 'cat_t2' and 'cat_t2_label'.

Each line in the data can be templated as a hypothesis as follows: "Cooperation is significantly 'comparative' when 'iv_label' is 'cat_t1_label' compared to when 'iv_label' is 'cat_t2_label'.". 'iv_label', 'cat_t1_label' and 'cat_t2_label' comes from the data ; 'comparative' must be added and can only take the values 'higher' or 'lower'.

Given the data, you must extract the subset of 5 rows that represent the most coherent hypotheses, ranked from the most coherent one. If there are less than 5 rows, you can re-order them by order of coherence. On top of the existing columns, you must add another one, 'comparative' that can take the values 'higher' or 'lower'.

The output must be in csv format.

Figure 10. Prompt used for the `reg` hypothesis.

D. User Studies

Figure 13 shows the distribution of users across *sivs* and *moderators*, distinguished by the type of hypothesis. There is more overlap for the common *sivs* and *moderators* used across users for the generated hypotheses, which shows that there would be a higher consensus for the *interesting siv* to explore for a hypothesis.

Figure 14 shows the distribution of the hypotheses' score across the type of hypothesis and models.

Prompt:

You will be given data in .csv format, where each line can be read as a hypothesis on how humans cooperate. The csv has the following columns: 'dependent', 'giv_prop', 'iv', 'iv_label', 'cat_t1', 'cat_t1_label', 'cat_t2', 'cat_t2_label', 'mod', 'mod_label', 'mod_val' and 'mod_val_label'.

Each line in the data can be templated as a hypothesis as follows: "When comparing studies where 'iv_label' is 'cat_t1_label' and studies where 'iv_label' is 'cat_t2_label', cooperation is significantly 'comparative' when 'mod_label' is 'mod_val_label' compared to when 'mod_label' has another value.". 'iv_label', 'cat_t1_label', 'cat_t2_label', 'mod_label', and 'mod_val_label' comes from the data ; 'comparative' must be added and can only take the values 'higher' or 'lower'.

Given the data, you must extract the subset of 5 rows that represent the most coherent hypotheses, ranked from the most coherent one. If there are less than 5 rows, you can re-order them by order of coherence. On top of the existing columns, you must add another one, 'comparative' that can take the values 'higher' or 'lower'.

The output must be in csv format.

Figure 11. Prompt used for the `study-mod` hypothesis.

Prompt:
You will be given data in .csv format, where each line can be read as a hypothesis on how humans cooperate. The csv has the following columns: 'dependent', 'giv_prop', 'iv', 'iv_label', 'cat_t1', 'cat_t1_label', 'cat_t2', 'cat_t2_label', 'mod', 'mod_label', 'mod_t1', 'mod_t1_label', 'mod_t2' and 'mod_t2_label'.
Each line in the data can be templated as a hypothesis as follows: "When comparing studies where 'iv_label' is 'cat_t1_label' and studies where 'iv_label' is 'cat_t2_label', cooperation from studies involving 'mod_t1_label' as 'mod_label' is significantly 'comparative' than cooperation from studies involving based on 'mod_t2_label' as 'mod_label'." 'iv_label', 'cat_t1_label', 'cat_t2_label', 'mod_label', 'mod_t1_label', and 'mod_t2_label' comes from the data ; 'comparative' must be added and can only take the values 'higher' or 'lower'.
Given the data, you must extract the subset of 5 rows that represent the most coherent hypotheses, ranked from the most coherent one. If there are less than 5 rows, you can re-order them by order of coherence. On top of the existing columns, you must add another one, 'comparative' that can take the values 'higher' or 'lower'.
The output must be in csv format.

Figure 12. Prompt used for the var-mod hypothesis.

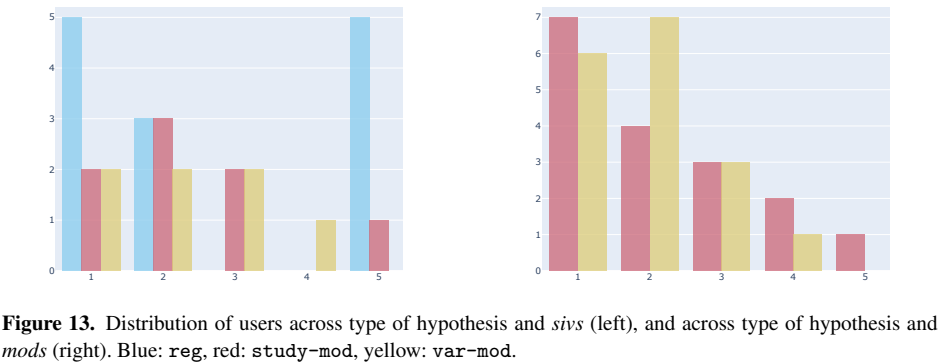


Figure 13. Distribution of users across type of hypothesis and *sivs* (left), and across type of hypothesis and *mods* (right). Blue: reg, red: study-mod, yellow: var-mod.

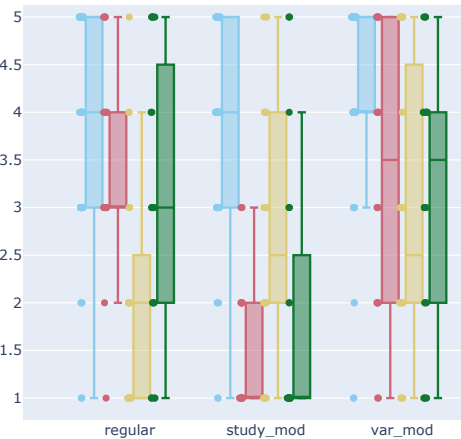


Figure 14. Distribution of hypotheses' score across type of hypothesis and models. Blue: human, green: AnyBURL, red: classification, yellow: LLM.